Exercise 6.9 SOLUTION

Q.1. Find the 9th term of the following harmonic sequences:

i. $\frac{1}{3},\frac{1}{5},\frac{1}{7},\ldots$

Sol: Given $\frac{1}{3},\frac{1}{5},\frac{1}{7},\ldots$ are in H.P, $a_9 = ?$

Then $\frac{3}{1},\frac{5}{1},\frac{7}{1},\ldots$ are in A.P

or 3,5,7..... are in A.P

$a_1 = 3, d = 5-3 = 2, n = 9$

$a_n = a_1 + (n-1)d$

$a_9 = 3+(9-1)2$

$= 3+(8)2 = 3+16 = 19$ in A.P $\Rightarrow a_9 = \frac{1}{19}$ in H.P.

نوٹ

سوال میں دی گئی ٹرم یا نمبر H.P میں ہیں جب ان کا reciprocal لیں گے تو وہ A.P میں چلے جائیں گے۔ اور جواب بھی A.P میں آجائے گا۔ آخر میں بھی جواب کا reciprocal لیں گے تو جواب H.P میں آجائے گا

ii. $\frac{-1}{5},\frac{-1}{3},-1,\ldots$

Sol: Given $\frac{-1}{5},\frac{-1}{3},-1$ are in H.P or $-5,-3,-1\ldots$ are in A.P

$a_1 = 5, d = -3-(-5) = -3+5 = 2, n = 9$

$a_n = a_1 + (n-1)d \Rightarrow a_9 = 5+(9-1)2$

$a_9 = -5+(8)(2) = -5+16 = 11$ in A.P then $a_9 = \frac{1}{11}$ in H.P

Q.2. Insert five harmonic means between the following given numbers:

i) $\frac{-2}{5}$ & $\frac{2}{13}$

Sol: Let $H_1, H_2, H_3, H_4, H_5,$ be five

H.M between $\frac{-2}{5}$ & $\frac{2}{13}$ then $\frac{-2}{5}, H_1, H_2, H_3, H_4, H_5, \frac{2}{13}$ are in H.P

CamScanner

$\Rightarrow \frac{-5}{2}, \frac{1}{H_1}, \frac{1}{H_2}, \frac{1}{H_3}, \frac{1}{H_4}, \frac{1}{H_5}, \frac{13}{2}$ are in A.P.

$a_1 = \frac{-5}{2}$ & $a_7 = a_1 + 6d = \frac{13}{2} \Rightarrow \frac{-5}{2} + 6d = \frac{13}{2} \Rightarrow 6d = \frac{13}{2} + \frac{5}{2} = \frac{18}{2} \Rightarrow d = \frac{18}{2} \times \frac{1}{6} \Rightarrow d = \frac{3}{2}$

$\frac{1}{H_1} = a_2 = a_1 + d = -\frac{5}{2} + \frac{3}{2} = \frac{-5+3}{2} = \frac{-2}{2} = -1$

$\frac{1}{H_2} = a_3 = a_1 + 2d = -\frac{5}{2} + 2\left(\frac{3}{2}\right) = -\frac{5}{2} + \frac{6}{2} = \frac{-5+6}{2} = \frac{1}{2}$

$\frac{1}{H_3} = a_4 = a_1 + 3d = -\frac{5}{2} + 3\left(\frac{3}{2}\right) = -\frac{5}{2} + \frac{9}{2} = \frac{-5+9}{2} = \frac{4}{2} = 2$

$\frac{1}{H_4} = a_5 = a_1 + 4d = -\frac{5}{2} + 4\left(\frac{3}{2}\right) = -\frac{5}{2} + \frac{12}{2} = \frac{-5+12}{2} = \frac{7}{2}$

$\frac{1}{H_5} = a_6 = a_1 + 5d = -\frac{5}{2} + 5\left(\frac{3}{2}\right) = -\frac{5}{2} + \frac{15}{2} = \frac{-5+15}{2} = \frac{10}{2} = 5$

Hence $H_1 = -1, H_2 = 2, H_3 = \frac{1}{2}, H_4 = \frac{2}{7}, H_5 = \frac{1}{5}$

ii) $\frac{1}{4}$ & $\frac{1}{24}$

Sol: Let H_1, H_2, H_3, H_4, H_5, H.M,s between $\frac{1}{4}$ & $\frac{1}{24}$ then

$\frac{1}{4}, H_1, H_2, H_3, H_4, H_5, \frac{1}{24}$ are in H.P

$\Rightarrow 4, \frac{1}{H_1}, \frac{1}{H_2}, \frac{1}{H_3}, \frac{1}{H_4}, \frac{1}{H_5}, 24$ are in A.P.

$a_1 = 4$ & $a_7 = a_1 + 6d = 24$

$4 + 6d = 24 \quad \Rightarrow \quad 6d = 24 - 4$

$\Rightarrow 6d = 24 - 4 = 20 \Rightarrow d = \frac{20}{6} = \frac{10}{3}$

Now

$\frac{1}{H_1} = a_2 = a_1 + d = 4 + \frac{10}{3} = \frac{12+10}{3} = \frac{22}{3}$

$\frac{1}{H_2} = a_3 = a_1 + 2d = 4 + 2\left(\frac{10}{3}\right) = \frac{12+20}{3} = \frac{32}{3}$

$\frac{1}{H_3} = a_4 = a_1 + 3d = 4 + 3\left(\frac{10}{3}\right) = \frac{12+30}{3} = \frac{42}{3} = 14$

$\frac{1}{H_4} = a_5 = a_1 + 4d = 4 + 4\left(\frac{10}{3}\right) = \frac{12+40}{3} = \frac{52}{3}$

$\frac{1}{H_5} = a_6 = a_1 + 5d = 4 + 5\left(\frac{10}{3}\right) = \frac{12+50}{3} = \frac{62}{3}$

Hence $H_1 = \frac{3}{22}, H_2 = \frac{3}{32}, H_3 = \frac{1}{14}, H_4 = \frac{3}{52}, H_5 = \frac{3}{62}$

CamScanner

Q.3. **The first term of an H.P. is $-\frac{1}{3}$ and the fifth term is $\frac{1}{5}$. Find its 9th term.**

Sol: $a_1 = -\frac{1}{3}, a_5 = \frac{1}{5}, a_9 = ?$ in H.P

$a_1 = -3$, $a_5 = 5$ in A.P _____(i)

$a_5 = a_1 + 4d = 5 \Rightarrow -3 + 4d = 5 \Rightarrow 4d = 8 \Rightarrow d = 2$ use (i)

$a_n = a_1 + (n-1)d$

$a_9 = -3 + (9-1)(2)$

$a_9 = -3 + (8)(2) = -3 + 16 = 13$

$a_9 = 13$ in A.P $\Rightarrow a_9 = \frac{1}{13}$ in H.P

Q.4. **If 5 is the harmonic mean between 2 & b .Find b .**

Sol: Given $a = 2, b = b, H.M = 5$

$H.M = \frac{2ab}{a+b}$

Put values $5 = \frac{2(2)b}{2+b} \Rightarrow 5 = \frac{4b}{2+b}$

$\Rightarrow 5(2+b) = 4b \Rightarrow 10 + 5b = 4b$

$\Rightarrow 10 + 5b - 4b = 0 \Rightarrow 10 + b = 0 \Rightarrow b = -10$

Q.5. **If the numbers $\frac{1}{k}, \frac{1}{2k+1}$ & $\frac{1}{4k-1}$ are in harmonic sequence, find k.**

Sol: $\frac{1}{k}, \frac{1}{2k+1}$ & $\frac{1}{4k-1}$ are in H.P

$k, 2k+1, 4k-1$ are in A.P

$\Rightarrow \quad 4k-1-(2k+1) = 2k+1-k$

$4k-1-2k-1 = k+1 \Rightarrow 2k-2 = k+1$

$\Rightarrow k-3 = 0 \Rightarrow k = 3$

Q.6. **Find n so that $\frac{a^{n+1} + b^{n+1}}{a^n + b^n}$ may be H.M. between a & b.**

Sol: If $\frac{a^{n+1} + b^{n+1}}{a^n + b^n}$ be H.M between a & b then

$\frac{a^{n+1} + b^{n+1}}{a^n + b^n} = \frac{2ab}{a+b}$

$(a^{n+1} + b^{n+1})(a+b) = (a^n + b^n)(2ab)$

$a^{n+2} + a^{n+1}b + ab^{n+1} + b^{n+2} = 2a^{n+1}b + 2ab^{n+1}$

$a^{n+2} + b^{n+2} = 2ab^{n+1} + 2ab^{n+1} - a^{n+1}b - ab^{n+1}$

$a^{n+2} + b^{n+2} = a^{n+1}b + ab^{n+1}$

$a^{n+2} - a^{n+1}b = ab^{n+1} - b^{n+2}$

$a^{n+1}(a-b) = b^{n+1}(a-b)$

$\frac{a^{n+1}}{b^{n+1}} = \frac{\cancel{(a-b)}}{\cancel{(a-b)}} = 1 \Rightarrow \left(\frac{a}{b}\right)^{n+1} = \left(\frac{a}{b}\right)^0 \Rightarrow n+1 = 0 \Rightarrow n = -1$ Note $1 = \left(\frac{a}{b}\right)^0$

CamScanner

Q.7. If a^2, b^2 & c^2 are in A.P., show that $a+b, c+a$ & $b+c$ are in H.P.

Sol: a^2, b^2 & c^2 are in A.P

Then $b^2 - a^2 = c^2 - b^2 \quad (i)$

We have to prove.

Now $a+b, c+a, b+c$ are in H.P

If $\frac{1}{a+b}, \frac{1}{c+a}, \frac{1}{b+c}$ are in A.P $\Rightarrow \frac{1}{b+c} - \frac{1}{c+a} = \frac{1}{c+a} - \frac{1}{a+b}$

$\Rightarrow \frac{(c+a)-(b+c)}{(b+c)(c+a)} = \frac{a+b-c-a}{(b+c)(c+a)} \Rightarrow \frac{\cancel{c}+a-b-\cancel{c}}{(b+c)(c+a)} = \frac{\cancel{a}+b-c-\cancel{a}}{(a+b)(c+a)}$

'$\times$' both sides by $(c+a)$

$\cancel{(c+a)} \frac{(a-b)}{(b+c)\cancel{(c+a)}} = \cancel{(c+a)} \frac{(b-c)}{(a+b)\cancel{(c+a)}} \quad \Rightarrow \frac{a-b}{b+c} = \frac{b-c}{a+b}$

By cross multiplication

$(a+b)(a-b) = (b+c)(b-c)$

$a^2 - b^2 = b^2 - c^2$

'$\times$' both sides by (-1)

$b^2 - a^2 = c^2 - b^2 \Rightarrow b^2 - a^2 = b^2 - a^2$ (use i)

Hence proved.

Q.8. If the H.M. and A.M. between two numbers are 4 and $\frac{9}{2}$ respectively, find the numbers.

Sol: $H.M = 4, A.M = \frac{9}{2}, a, b = ?$

$H.M = \frac{2ab}{a+b} \Rightarrow 4 = \frac{2ab}{a+b}$ _____ (i)

$A.M = \frac{a+b}{2} \Rightarrow \frac{9}{2} = \frac{a+b}{2}$

$\Rightarrow a+b = 9$ _____ (ii)

Put equation (ii) in (i) $\Rightarrow 4 = \frac{2ab}{9} \Rightarrow 2ab = 36$

$\Rightarrow ab = \frac{36}{2} = 18 \Rightarrow ab = 18$ _____ (iii)

from (ii) $\Rightarrow a+b = 9 \Rightarrow a = 9-b$

Put value in (iii)

$(9-b)b = 18 \Rightarrow 9b - b^2 - 18 = 0$

$\Rightarrow b^2 - 9b + 18 = 0$

$\Rightarrow b^2 - 3b - 6b + 18 = 0$

$\Rightarrow b(b-3) - 6(b-3) = 0$

$\Rightarrow b-3 = 0 \text{ or } b-6 = 0 \Rightarrow b = 3 \text{ or } b = 6$

When $b = 3$ *then* $a = 9-3 = 6$

When $b = 6$ *then* $a = 9-6 = 3$

Numbers are $6,3$ *or* $3,6$

CamScanner

Q.9. **If the (positive) G.M. and H.M. between two numbers are 4 and $\frac{16}{5}$ find the numbers.**

Sol: $G.M=4, H.M=\frac{16}{5}, a,b=?$

$G.M=\sqrt{ab} \Rightarrow 4=\sqrt{ab} \Rightarrow ab=16 \quad (i)$

$H.M=\frac{2ab}{a+b} \Rightarrow \frac{16}{5}=\frac{2ab}{a+b}$

$\Rightarrow \frac{16}{5}=\frac{2(16)}{a+b} \Rightarrow \frac{1}{5}=\frac{2}{a+b} \Rightarrow a+b=10 \quad (ii)$

From (ii) $a=10-b \quad (iii)$

Eq. (i) Becomes $(10-b)b=16 \Rightarrow 10b-b^2-16=0 \Rightarrow b^2-10b+16=0$

$\Rightarrow b^2-2b-8b+16=0$

$\Rightarrow b(b-2)-8(b-2)=0$

$\Rightarrow (b-2)(b-8)=0$

$\Rightarrow b-2=0$ or $b-8=0$

$\Rightarrow b=2$ or $b=8$

When $b=2$ then $a=10-2=8$

When $b=8$ then $a=10-8=2$

Numbers are $8,2$ or $2,8$

Q.10. **If $\frac{b+c-a}{a}, \frac{c+a-b}{b}, \frac{a+b-c}{c}$ are in A.P., show that a,b,c are in H.P.**

Sol: Given that $\frac{b+c-a}{a}, \frac{c+a-b}{b}, \frac{a+b-c}{c}$ are in AP

i.e. Common difference remain constant

$\frac{c+a-b}{b}-\frac{b+c-a}{a}=\frac{a+b-c}{c}-\frac{c+a-b}{b}$

$2\left(\frac{c+a-b}{b}\right)=\frac{b+c-a}{a}+\frac{a+b-c}{c}$

Multiplying both sides by abc

$2ac(c+a-b)=bc(b+c-a)+ab(a+b-c)$

$2ac^2+2a^2c-2abc=b^2c+bc^2-abc+a^2b+ab^2-abc$

$=b^2c+bc^2+a^2b+ab^2-2abc$

$2ac^2+2a^2c=a^2b+ab^2+b^2c+bc^2$

$2ac(a+c)=ab(a+b)+bc(b+c)$

Divide by abc: $\frac{2(a+c)}{b}=\frac{a+b}{c}+\frac{b+c}{a}$

$\frac{2a}{b}+\frac{2c}{b}=\frac{a}{c}+\frac{b}{c}+\frac{b}{a}+\frac{c}{a}$

$\frac{2}{b}=\frac{1}{a}+\frac{1}{c} \Rightarrow \frac{1}{b}+\frac{1}{b}=\frac{1}{a}+\frac{1}{c} \Rightarrow \frac{1}{b}-\frac{1}{a}=\frac{1}{c}-\frac{1}{b}$

It means $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P. therefor a, b, c are in H.P.

2nd Method

Given

$\frac{b+c-a}{a}, \frac{c+a-b}{b}, \frac{a+b-c}{c}$ are in A.P

Adding 2 in each term

$\frac{b+c-a}{a}+2, \frac{c+a-b}{b}+2, \frac{a+b-c}{c}+2$ are in A.P

$\frac{b+c-a+2a}{a}, \frac{c+a-b+2b}{b}, \frac{a+b-c+2c}{c}$ are in A.P

$\frac{a+b+c}{a}, \frac{a+b+c}{b}, \frac{a+b+c}{c}$ are in A.P

Divide each term by $(a+b+c)$

$\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P

Hence a,b,c are in H.P

CamScanner

Q.11. If a,b,c,d are in H.P., show that $3(a-b)(c-d)=(b-c)(a-d)$.

Sol: Given that a,b,c,d are in HP.

Then $\frac{1}{a},\frac{1}{b},\frac{1}{c},\frac{1}{d}$ are in AP.

Consider

$\frac{1}{a}=x,\frac{1}{b}=x+k,\frac{1}{c}=x+2k,\frac{1}{d}=x+3k$

Then

$a=\frac{1}{x},b=\frac{1}{x+k},c=\frac{1}{x+2k},d=\frac{1}{x+3k}$

we have to prove: $3(a-b)(c-d)=(b-c)(a-d)$

L.H.S$=3(a-b)(c-d)$

$=3\left(\frac{1}{x}-\frac{1}{x+k}\right)\left(\frac{1}{x+2k}-\frac{1}{x+3k}\right)=3\left(\frac{x+k-x}{x(x+k)}\right)\left(\frac{x+3k-x-2k}{(x+2k)(x+3k)}\right)=3\left(\frac{k}{x(x+k)}\right)\left(\frac{k}{(x+2k)(x+3k)}\right)$

$=\frac{3k^2}{x(x+k)(x+2k)(x+3k)}$ (i)

R.H.S$=(b-c)(a-d)$

$=\left(\frac{1}{x+k}-\frac{1}{x+2k}\right)\left(\frac{1}{x}-\frac{1}{x+3k}\right)=\left(\frac{x+2k-x-k}{(x+k)(x+2k)}\right)\left(\frac{x+3k-x}{x(x+3k)}\right)=\left(\frac{k}{(x+k)(x+2k)}\right)\left(\frac{3k}{x(x+3k)}\right)$

$=\frac{3k^2}{x(x+k)(x+2k)(x+3k)}$ (ii)

From (i) &(ii): $\Rightarrow L.H.S=R.H.S$ or $3(a-b)(c-d)=(b-c)(a-d)$

Q.12. If between any two numbers there are inserted two A.M., A_1,A_2, two G.Ms G_1, G_2 and two H.Ms. H_1, H_2, show that $\frac{A_1+A_2}{G_1G_2}=\frac{H_1+H_2}{H_1H_2}$.

Sol: If A_1 & A_2 are two arithmetic means inserted between a & b, then a,A_1,A_2,b are in A.P.

Here $a_1=a$ and $a_4=a_1+3d=b\Rightarrow d=\frac{b-a}{3}$

$A_1=a+d$

So, $A_1=a+\frac{b-a}{3}=\frac{3a+b-a}{3}=\frac{2a+b}{3}$

$A_2=a+2d$

$A_2=a+2\left(\frac{b-a}{3}\right)=\frac{3a+2b-2a}{3}=\frac{a+2b}{3}$

Therefore, $A_1+A_2=\frac{2a+b}{3}+\frac{a+2b}{3}=\frac{3a+3b}{3}=a+b$.

If G_1 & G_2 are two geometric means inserted between a & b, then a,G_1,G_2,b are in G.P.

Let the common ratio be r.

$G_1=ar$

$G_2=ar^2$

$b=ar^3\Rightarrow r^3=\frac{b}{a}\Rightarrow r=\left(\frac{b}{a}\right)^{1/3}$

CamScanner

So, $G_1 = a\left(\frac{b}{a}\right)^{1/3} = a^{2/3}b^{1/3}$

$G_2 = a\left(\frac{b}{a}\right)^{2/3} = a^{1/3}b^{2/3}$

Therefore, $G_1.G_2 = \left(a^{2/3}b^{1/3}\right)\left(a^{1/3}b^{2/3}\right) = a^{2/3+1/3}b^{1/3+2/3} = a^1b^1 = ab.$

If H_1 & H_2 are two harmonic means inserted between a & b, then a, H_1, H_2, b are in H.P., which means $\frac{1}{a}, \frac{1}{H_1}, \frac{1}{H_2}, \frac{1}{b}$ are in A.P. Let the common difference be d'.

$\frac{1}{H_1} = \frac{1}{a} + d'$

$\frac{1}{H_2} = \frac{1}{a} + 2d'$

$\frac{1}{H_3} = \frac{1}{a} + 3d' \quad \Rightarrow \quad 3d' = \frac{1}{b} - \frac{1}{a} = \frac{a-b}{ab} \quad \Rightarrow \quad d' = \frac{a-b}{3ab}$

So, $\frac{1}{H_1} = \frac{1}{a} + \frac{a-b}{3ab} = \frac{3b+a-b}{3ab} = \frac{a+2b}{3ab} \quad \Rightarrow \quad H_1 = \frac{3ab}{a+2b}$

$\frac{1}{H_2} = \frac{1}{a} + 2\left(\frac{a-b}{3ab}\right) = \frac{3b+2a-2b}{3ab} = \frac{2a+b}{3ab} \quad \Rightarrow \quad H_2 = \frac{3ab}{2a+b}$

Therefore, $H_1 + H_2 = \frac{3ab}{a+2b} + \frac{3ab}{2a+b} = 3ab\left(\frac{1}{a+2b} + \frac{1}{2a+b}\right) = 3ab\left(\frac{2a+b+a+2b}{(a+2b)(2a+b)}\right)$

$= 3ab\left(\frac{3a+3b}{2a^2+ab+4ab+2b^2}\right) = \frac{9ab(a+b)}{2a^2+5ab+2b^2}$

And, $H_1.H_2 = \left(\frac{3ab}{a+2b}\right)\left(\frac{3ab}{2a+b}\right) = \frac{9a^2b^2}{(a+2b)(2a+b)} = \frac{9a^2b^2}{2a^2+5ab+2b^2}$

$\Rightarrow \frac{H_1+H_2}{H_1.H_2} = \frac{\frac{9ab(a+b)}{2a^2+5ab+2b^2}}{\frac{9a^2b^2}{2a^2+5ab+2b^2}} = \frac{\cancel{9ab}(a+b)}{\cancel{2a^2+5ab+2b^2}} \times \frac{\cancel{2a^2+5ab+2b^2}}{\cancel{9a^2b^2}} = \frac{a+b}{ab} = \frac{A_1+A_2}{G_1G_2}$ Hence Proved.

Q.13 **The H.M. of two numbers is 4. The A.M., A and the G.M., G satisfy the relation $2A + G^2 = 27$. Find the numbers.**

Sol: Let the two numbers be a and b.

The harmonic mean (H.M.) is given by: $H = \frac{2ab}{a+b}$

We are given $H = 4$, so $\frac{2ab}{a+b} = 4 \Rightarrow 2ab = 4(a+b) \Rightarrow ab = 2(a+b)$

The geometric mean (G.M.) is given by: $G = \sqrt{ab}$, so $G^2 = ab$

We are given the relation $2A + G^2 = 27$.

Substituting the expressions for A and G^2:

$\cancel{2}\left(\frac{a+b}{\cancel{2}}\right) + ab = 27 \quad \Rightarrow a+b+ab = 27$

CamScanner

Now we have a system of two equations with two variables:

$ab = 2(a+b)$ .....(i)

$a+b+ab = 27$ .....(ii)

$(a+b)+2(a+b) = 27$ Put (i) in (ii)

$3(a+b) = 27$

$a+b = 9 \Rightarrow a = 9-b$ .....(iii)

Now substitute the value of $a+b$ back into eq (i):

$ab = 2(9)$

$ab = 18$ .....(iv)

Put value of $a = 9-b$ in (iv)

$(9-b)b = 18 \Rightarrow 9b-b^2 = 18$

$b^2-9b+18 = 0$

$b^2-3b-6b+18 = 0$

$b(b-3)-6(b-3) = 0$

$(b-3)(b-6) = 0$

$b-3 = 0$ as $b-6 = 0$

$a = 9-b$ or $a = 9-b$

$a = 9-3 = b$ or $a = 9-6 = 3$

Two numbers are 3,6 or 6,3

Q.14 **First three of the four numbers a, b, c, d are in A.P., and the next three are in H.P., show that $ad = bc$.**

Sol: Given that a, b, c are in arithmetic progression (A.P.), we have:

$b-a = c-b$

$2b = a+c \quad (i)$

Given that b, c, d are in harmonic progression (H.P.), their reciprocals $\frac{1}{b}, \frac{1}{c}, \frac{1}{d}$ are in arithmetic progre

Thus: $\frac{1}{c}-\frac{1}{b} = \frac{1}{d}-\frac{1}{c}$

$\frac{b-c}{bc} = \frac{c-d}{cd}$

$cd(b-c) = bc(c-d)$

$d(b-c) = b(c-d)$

$db-dc = bc-bd$

$2bd = bc+dc \quad (ii)$

From equation (i), we have $c = 2b-a$ .....(iii)

Substitute this into equation (ii):

$2bd = b(2b-a)+d(2b-a)$

$2bd = 2b^2-ab+2bd-ad \Rightarrow 2b^2-ab+\cancel{2bd}-ad-\cancel{2bd} = 0$

$ad = 2b^2-ab$

$ad = b(2b-a)$

Put $2b-a = c$ we get $ad = bc$

CamScanner

Q.15 If a,b,c are in G.P., show that $\log_a x, \log_b x, \log_c x$ are in H.P.

Sol:

Given that a,b,c are in G.P.

$$\frac{b}{a}=\frac{c}{b}$$

$$\Rightarrow b^2 = ac \text{ ---------(i)}$$

We have to show $\log_a x, \log_b x, \log_c x$ from an HP.

It is enough to show that $\frac{1}{\log_a x}, \frac{1}{\log_b x}, \frac{1}{\log_c x}$ from an AP.

$$\frac{1}{\log_b x}-\frac{1}{\log_a x}=\frac{1}{\log_c x}-\frac{1}{\log_b x}$$

$$\frac{\log b}{\log x}-\frac{\log a}{\log x}=\frac{\log c}{\log x}-\frac{\log b}{\log x} \Rightarrow \frac{\log b-\log a}{\cancel{\log x}}=\frac{\log c-\log b}{\cancel{\log x}}$$

$$\log b-\log a=\log c-\log b$$

$$\log\frac{b}{a}=\log\frac{c}{b}$$

$$\frac{b}{a}=\frac{c}{b}$$

$$b^2=ac \Rightarrow b^2=b^2 \text{ (Use I)}$$

Common difference is same so $\frac{1}{\log_{a^x}}, \frac{1}{\log_{b^x}}$ *and* $\frac{1}{\log_{c^x}}$ are in A.P. hence $\log_{a^x}, \log_{b^x}, \log_{c^x}$ are in H.P.

Q.16 If a,b,c are in H.P., show that

i) $\frac{a-b}{b-c}=\frac{a}{c}$ ii) $(a-c)^2=(a+c)(a-2b+c)$

Sol:

i) If a,b,c are in H.P., then $\frac{1}{a},\frac{1}{b},\frac{1}{c}$ are in A.P., which implies:

$$\frac{1}{b}-\frac{1}{a}=\frac{1}{c}-\frac{1}{b}$$

$$\frac{a-b}{ab}=\frac{b-c}{bc}$$

$$\cancel{ab}c\left(\frac{a-b}{\cancel{ab}}\right)=a\cancel{bc}\left(\frac{b-c}{a\cancel{bc}}\right)$$

$$c(a-b)=a(b-c)$$

$$\Rightarrow \frac{a-b}{b-c}=\frac{a}{c} \text{ hence proved.}$$

ii) To show $(a-c)^2=(a+c)(a-2b+c)$

Given a, b, c are in H.P then $\frac{1}{a},\frac{1}{b},\frac{1}{c}$ are in A.P

$$\frac{1}{c}-\frac{1}{b}=\frac{1}{b}-\frac{1}{a} \Rightarrow \frac{1}{b}+\frac{1}{b}=\frac{1}{c}+\frac{1}{a}$$

$$\frac{2}{b}=\frac{a+c}{ac} \Rightarrow \frac{b}{2}=\frac{ac}{a+c} \Rightarrow b=\frac{2ac}{a+c}$$

CamScanner

So $b=\frac{2ac}{a+c}$

R.H.S. $=(a+c)(a-2b+c)=(a+c)\left(a-2\left(\frac{2ac}{a+c}\right)+c\right)=(a+c)\left(\frac{a(a+c)-4ac+c(a+c)}{a+c}\right)$

$=a^2+ac-4ac+ca+c^2=a^2-2ac+c^2=(a-c)^2=L.H.S$

Hence Proved that $(a-c)^2=(a+c)(a-2b+c)$

Q.17 **If $2+x,5+x$ & $9+x$ are in H.P., find the value of x.**

Sol: Given that $2+x,5+x,\&9+x$ are in H.P., their reciprocals are in A.P.:

$\frac{1}{2+x},\frac{1}{5+x},\frac{1}{9+x}$ are in A.P.

Therefore:

$\frac{1}{5+x}-\frac{1}{2+x}=\frac{1}{9+x}-\frac{1}{5+x}$

$\frac{(2+x)-(5+x)}{(5+x)(2+x)}=\frac{(5+x)-(9+x)}{(9+x)(5+x)} \Rightarrow \frac{2+\cancel{x}-5-\cancel{x}}{2+x}=\frac{5+\cancel{x}-9-x}{9+x}$

$\frac{-3}{(2+x)}=\frac{-4}{(9+x)}$

$-3(9+x)=-4(2+x) \Rightarrow -27-3x=-8-4x$

$-3x+4x=-8+27 \Rightarrow x=19$

Q.18 **If the roots of the equation $a(b-c)x^2+b(c-a)x+c(a-b)=0$ are equal, prove that a,b,c are in H.P.**

Sol: **Given $a(b-c)x^2+b(c-a)x+c(a-b)=0$ -------- (i) has equal roots**

If we put $x=1$ which satisfies (i) and roots are equal so both roots are 1, 1

Product of roots $=\alpha\beta=\frac{C}{A}$

$$\Rightarrow(1)(1)=\frac{c(a-b)}{a(b-c)}$$

Product of roots: $\frac{c(a-b)}{a(b-c)}=1$

$c(a-b)=a(b-c)$

$ca-cb=ab-ac$

$2ac=ab+cb$

Dividing by abc:

$\frac{2ac}{abc}=\frac{ab}{abc}+\frac{cb}{abc}$

$\frac{2}{b}=\frac{1}{c}+\frac{1}{a} \Rightarrow \frac{1}{b}+\frac{1}{b}=\frac{1}{c}+\frac{1}{a} \Rightarrow \frac{1}{c}-\frac{1}{b}=\frac{1}{b}-\frac{1}{a}$ Which shows that $\frac{1}{a},\frac{1}{b},\frac{1}{c}$ are in A.P.

Hence a,b,c are in H.P.

CamScanner